A LOCAL FACTORISATION CRITERION FOR A MAP FOLLOWED BY A SURJECTION ONTO A FLAT MODULE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The sheaf of ideals generated by finitely many local functions, and the interpretation of the sum of the tensor products of those functions with the given sections as a section of the tensor product with the ideal sheaf. Used at the start of the counted proof (source lines 2993-2996). Turns the hypothesis that the displayed complex is a complex into a relation in a tensor product.
- The injectivity of the maps obtained by tensoring with a flat module, and the description of the tensor product of the ideal sheaf with a flat module by generators and relations. Used inside the counted proof (source lines 2996-3004), the last step by a cross-chapter pointer. Lifts the vanishing of the section of the tensor product to an explicit relation between the local functions and the local sections.

Source reference. Original reference algebra-lemma-relations: sources/stacks-modules/pinned-original-source.tex, source lines 3004–3004. Navigation pointer only: the counted text refers to the Stacks item algebra-lemma-relations, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let $(X, \mathcal{O}_X)$ be a ringed space. Let $\mathcal{F}$ be a flat $\mathcal{O}_X$-module. Let $U \subset X$ be open and let*

$$\mathcal{O}_U \xrightarrow{(f_1,\ldots,f_n)} \mathcal{O}_U^{\oplus n} \xrightarrow{(s_1,\ldots,s_n)} \mathcal{F}|_U$$

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be a complex of $\mathcal{O}_U$-modules. For every $x \in U$ there exists an open neighbourhood $V \subset U$ of x and a factorization

$$\mathcal{O}_V^{\oplus n} \xrightarrow{A} \mathcal{O}_V^{\oplus m} \xrightarrow{(t_1,\ldots,t_m)} \mathcal{F}|_V$$

of $(s_1, \ldots, s_n)|_V$ such that $A \circ (f_1, \ldots, f_n)|_V = 0$.

Proof. Let $\mathcal{I} \subset \mathcal{O}_U$ be the sheaf of ideals generated by $f_1, \ldots, f_n$. Then $\sum f_i \otimes s_i$ is a section of $\mathcal{I} \otimes_{\mathcal{O}_U} \mathcal{F}|_U$ which maps to zero in $\mathcal{F}|_U$. As $\mathcal{F}|_U$ is flat the map $\mathcal{I} \otimes_{\mathcal{O}_U} \mathcal{F}|_U \to \mathcal{F}|_U$ is injective. Since $\mathcal{I} \otimes_{\mathcal{O}_U} \mathcal{F}|_U$ is the sheaf associated to the presheaf tensor product, we see there exists an open neighbourhood $V \subset U$ of x such that $\sum f_i|_V \otimes s_i|_V$ is zero in $\mathcal{I}(V) \otimes_{\mathcal{O}(V)} \mathcal{F}(V)$. Unwinding the definitions using Algebra, Lemma algebra-lemma-relations we find $t_1, \ldots, t_m \in \mathcal{F}(V)$ and $a_{ij} \in \mathcal{O}(V)$ such that $\sum a_{ij} f_i|_V = 0$ and $s_i|_V = \sum a_{ij} t_j$. □